



MASENO UNIVERSITY
UNIVERSITY EXAMINATIONS 2024/2025

**FIRST YEAR SECOND SEMESTER EXAMINATIONS FOR THE
DEGREE OF BACHELOR OF SCIENCE WITH INFORMATION
TECHNOLOGY**

MAIN CAMPUS

MMA 114: DISCRETE MATHEMATICS I

Date: 8th May, 2025

Time: 3. 30 - 6. 30 pm

INSTRUCTIONS:

- DO NOT write anywhere on this question paper
- Answer question ONE (COMPULSORY) and any other TWO questions
- Proofs should be written carefully
- Observe further instructions on the answer booklet

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MMA 114: DISCRETE MATHEMATICS I

QUESTION ONE (Compulsory)

[30 Marks]

- (a) We want to design a flag that consists of three horizontal stripes; the colour of the middle stripe should be different from the other two stripes. How many possibilities are there, if the colours red, green, blue, yellow, black and white can be used? [3 Marks]
- (b) Provide a combinatorial proof of the following identity:
$$\binom{n+m}{r} = \sum_{k=0}^r \binom{n}{k} \binom{m}{r-k}.$$
 [4 Marks]
- (c) Consider the sequence $a_0, a_1, a_2, \dots$ defined by the initial conditions $a_0 = 2, a_1 = 1$, and the recurrence relation $a_n = a_{n-1} + a_{n-2}$ for $n = 2, 3, 4, \dots$. Find an explicit formula for a_n . [4 Marks]
- (d) Use generating functions to solve the recurrence relation $T_n = 2T_{n-1} + 1$ for $n = 1, 2, 3, \dots$. The initial condition is $T_0 = 0$. [4 Marks]
- (e) How many solutions does the equation $x_1 + x_2 + x_3 = 11$ have, where x_1, x_2 and x_3 are nonnegative integers? [3 Marks]
- (f) Show that it is impossible to place five points inside an equilateral triangle of sides 2 cm such that any two points are at least 1 cm apart. [3 Marks]
- (g) Find the binary expansion of $(591)_{10}$. [3 Marks]
- (h) Show that 229 is prime. [3 Marks]
- (i) Use Euclidean Algorithm to find the greatest common divisor of 9999 and 2007. [3 Marks]

QUESTION TWO

[20 Marks]

- (a) Find the prime factor decomposition of 144144. [2 Marks]
- (b) Use Fermat's little theorem to compute $5^{987654321} \pmod{7}$. [3 Marks]
- (c) (i) Use the Euclidean Algorithm to show that 21 and 71 are relatively prime. [3 Marks]
(ii) Find the inverse of 21 modulo 71. [3 Marks]
(iii) Find all solutions to the linear congruence: $21x \equiv 5 \pmod{71}$. [4 Marks]
- (d) Find the smallest integer having the remainders 3, 11, 15 when divided by 10, 13, 17, respectively. [5 Marks]

QUESTION THREE

[20 Marks]

- (a) Show that in any gathering of six people, there are either three people who are mutual friends or three people who are mutual enemies. [5 Marks]
- (b) Consider bit strings of length 12. How many of these strings have:
- (i) exactly four 0's? [2 Marks]
 - (ii) at least four 0's? [3 Marks]
 - (iii) an odd number of 1's? [2 Marks]
 - (iv) alternating 0's and 1's? [2 Marks]
 - (v) four consecutive 0's? [3 Marks]
 - (vi) no consecutive 0's? [3 Marks]

QUESTION FOUR

[20 Marks]

- (a) How many different number plates for cars can be made if each number plate contains one letter (A-Z) followed by three digits (2-9) and two letters (A-Z) except I and O? [4 Marks]
- (b) A sequence $P_0, P_1, \dots$ is given by the recurrence relation: $P_n = 8P_{n-1} + 10^{n-1}$ with initial condition $P_0 = 1$. Use generating functions to obtain the explicit formula for P_n . [7 Marks]
- (c) All the houses on one side of a certain street are to be painted either green or blue. In how many ways can this be done if there are n houses and there may not be two green houses next to each other? [6 Marks]
- (d) What is the coefficient of $w^3x^6y^5z^7$ in the expansion of $(2w - 3x + y - z)^{21}$? [3 Marks]

QUESTION FIVE

[20 Marks]

- (a) In how many possible orders can the letters of the word MISSISSIPPI be arranged? [2 Marks]
- (b) How many numbers between 1 and 1000000 are neither a square nor a cube of an integer? [4 Marks]
- (c) Suppose that a person deposits ten thousand shillings in a savings account at a bank yielding 15% per year with interest compounded annually. Use iteration method to find the amount that will be in the account after 20 years, if no withdrawal was ever done? [4 Marks]
- (d) State the *Binomial Theorem* and hence use it to prove the following identity:
$$\sum_{k=0}^n \binom{n}{k} 2^k = 3^n.$$
 [3 Marks]
- (e) Provide a combinatorial proof of the identity in (d) above. [4 Marks]
- (f) Use a combinatorial reasoning (not binomial theorem) to find the expansion of $(x + y)^4$. [3 Marks]